

SAYERI BISWAS

AI Security | LLM Security | AI Red Teaming | Adversarial ML | Python

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PROFESSIONAL SUMMARY

Technology Analyst with nearly 5 years at Infosys across secure engineering, application security and AI security, with current experience in AI security evaluation, adversarial testing, LLM security and secure AI engineering. Experience evaluating security risks across LLM applications, RAG pipelines, AI agents, ML models and AI deployment pipelines, including prompt injection, jailbreaks, data poisoning, model robustness, insecure tool execution and sensitive-information disclosure.

Develops Python-based security evaluation utilities and automated test pipelines for AI systems, analyzes model behavior and security evidence, validates defensive controls, and integrates AI security testing into CI/CD and LLMOps workflows. Experienced in working with cybersecurity, data science, Responsible AI, platform and engineering teams on AI security experiments, technical research, remediation and pre-deployment security validation.

CORE EXPERTISE

AI Security: AI Security Evaluation, Adversarial Testing, AI Red Teaming, AI Threat Modeling, AI Vulnerability Analysis, Secure AI Engineering, AI Security Research, Security Testing

LLM Security: Prompt Injection, Indirect Prompt Injection, Jailbreaks, Sensitive Information Disclosure, System Prompt Leakage, Insecure Output Handling, Excessive Agency, Data Exfiltration

RAG & Agent Security: RAG Security, Retrieval Security, Context Poisoning, Malicious Documents, Retrieval Manipulation, Document Authorization, Tool Authorization, Excessive Agency, Agent Hijacking, Memory Security, MCP Security, Human Approval

ML Security: Adversarial ML, Evasion, Data Poisoning, Model Poisoning, Model Robustness, Model Extraction, Model Inversion, Membership Inference, Training-Data Security

MLOps / LLMOps: Model Evaluation, Security Regression Testing, Model Deployment, CI/CD, Automated Evaluation, Model Monitoring, Security Telemetry, Model Versioning

TECHNICAL SKILLS

Programming: Python, SQL, Bash

AI / ML: Machine Learning, Deep Learning, LLMs, RAG, Embeddings, Vector Search, PyTorch, LangChain, LangGraph, AI Agents, Multi-Agent Systems

AI Security Tooling: PyRIT, TextAttack, IBM Adversarial Robustness Toolbox (ART), MITRE ATLAS

Application Security: API Security, Authentication, Authorization, OAuth 2.0, OIDC, RBAC, Secure Coding, Vulnerability Assessment, Secure SDLC

Cloud / Platform: Azure, Docker, REST APIs, Secrets Management, Logging, Monitoring

DevSecOps: Git, Jenkins, CI/CD, SAST, Dependency Scanning, Security Regression Testing

PROFESSIONAL EXPERIENCE

INFOSYS LIMITED

Technology Analyst

December 2023 - Present

AI Security | Generative AI | Adversarial Evaluation

- Conduct security evaluations of enterprise LLM applications, RAG systems, AI agents, ML inference services and AI deployment pipelines across model, data, application and tool layers.
- Design adversarial security tests covering prompt injection, indirect prompt injection, jailbreaks, sensitive-information disclosure, excessive agency, insecure tool execution and data-exfiltration scenarios.
- Build Python-based security testing utilities for test generation, evaluation execution, response capture, application telemetry and structured security evidence.
- Analyze model responses, application logs, retrieved context and tool-execution traces to identify unsafe behavior, security-control failures and attack paths; evaluate RAG security for malicious documents, context poisoning, retrieval manipulation, unauthorized retrieval and sensitive-data exposure.
- Test agentic workflows for tool authorization, excessive agency, unsafe invocation, memory/context manipulation and human approval; evaluate MCP authorization boundaries and permissions.
- Develop security regression tests and integrate AI security evaluations into CI/CD and LLMOps; support ML robustness experiments involving adversarial examples, evasion and data poisoning using Python and PyTorch.
- Support AI security research and pre-deployment reviews by validating security controls, configuration, access boundaries, guardrails, dependencies and logging; prepare evidence-based findings.

System Engineer

January 2022 - November 2023

Secure Engineering | Application Security | AI/ML Platform Support

- Supported secure software development across requirements analysis, design, implementation, testing, deployment and production support; performed application and API security reviews covering authentication, authorization, input validation and sensitive-data handling.
- Developed Python automation for API validation, application testing, data-processing workflows and engineering utilities; supported vulnerability assessment, remediation, regression testing and closure evidence.
- Implemented secure coding controls around input validation, exception handling, access checks, secrets handling and API communication.
- Supported CI/CD testing, release validation and deployment checks; contributed to ML/AI-enabled enterprise application development and developed foundational understanding of model inputs, outputs, evaluation and data-processing pipelines.
- Participated in production troubleshooting, root-cause analysis and post-deployment validation.

PROJECTS

Enterprise LLM Security Evaluation Framework

Python / PyRIT / TextAttack / MITRE ATLAS / FastAPI

- Built reusable Python test harnesses for adversarial prompt generation, security evaluation, evidence capture and security regression testing for LLM applications.
- Designed repeatable evaluation workflows covering direct prompt injection, indirect prompt injection, jailbreaks, sensitive-information disclosure and unsafe model outputs.
- Generated structured adversarial test cases and executed evaluation workflows across model and application behavior.
- Analyzed model responses, application behavior and security-control results to identify unsafe behavior, control failures and potential attack paths.
- Captured prompts, responses, application behavior and security-control results as structured security evidence for analysis and reporting.
- Mapped selected attack scenarios to MITRE ATLAS techniques and documented attack assumptions, observations and defensive measures.
- Added regression tests to detect recurrence after prompt, model, application or guardrail changes.

Secure RAG & Agent Security Lab

Python / LangGraph / RAG / Vector Search / MCP / Docker

- Developed an isolated environment to study security risks across retrieval, context handling and agent-execution layers.
- Tested malicious documents, context poisoning, retrieval manipulation and unauthorized retrieval scenarios within controlled environments.
- Implemented document-level access checks and metadata-based filtering to validate retrieval authorization behavior.
- Evaluated retrieval behavior for potentially malicious or unauthorized content and analyzed security implications of retrieved context.
- Tested excessive agency, unauthorized tool execution and malicious tool inputs across agent workflows.
- Evaluated MCP-connected tool boundaries, authorization controls and controlled execution of downstream enterprise actions.
- Tested human approval controls for high-impact agent actions and documented security observations.
- Added audit logging for retrieval activity, agent decisions and downstream tool calls to support security analysis.

Adversarial ML Evaluation

Python / PyTorch / IBM ART

- Developed controlled experiments to understand model robustness and adversarial machine-learning attack techniques.
- Implemented adversarial examples and evaluated model behavior under controlled evasion conditions.
- Simulated controlled data-poisoning scenarios and evaluated model-robustness behavior and adversarial input detection.
- Used Python and PyTorch to support controlled experiments involving adversarial inputs and model behavior.
- Studied model extraction, model inversion and membership-inference attack concepts in isolated environments.
- Documented attack assumptions, observations, defensive measures, evaluation results and limitations.

AI SECURITY FRAMEWORKS

- **OWASP:** OWASP Top 10 for LLM Applications; OWASP Top 10 for Agentic Applications.
- **NIST:** NIST AI Risk Management Framework; NIST Generative AI Profile; NIST Adversarial Machine Learning Taxonomy.
- **MITRE:** MITRE ATLAS; MITRE ATT&CK.
- **Secure Development:** Secure Software Development Framework (SSDF).

CERTIFICATIONS

- Infosys Certified Responsible AI Foundation
- Infosys Certified Penetration Testing Expert
- Infosys Certified GenerativeAI Professional
- Infosys Certified Python Associate

EDUCATION

- **Bachelor of Engineering - Electrical & Engineering** 2020
- RCC Institute of Information Technology

LANGUAGES

- English - Professional Proficiency
- Hindi - Professional Proficiency
- Bengali - Native Proficiency
- Japanese - Beginner